

Second Quantization

Single - Particle operators:

$$\hat{A} = \sum_{n=1}^N A(q_n, p_n)$$

$$\hat{A} | \dots n_\mu \dots \rangle = \sum_{\mu} A_{\mu\mu} n_{\mu} | \dots n_{\mu} \dots \rangle$$

$$= \sum_{\mu} A_{\mu\mu} a_{\mu}^+ a_{\mu}^- | \dots n_{\mu} \dots \rangle$$

change of basis . $a_{\mu}^- = \sum_n C_{\mu n}^* a_n^-$; $a_{\mu}^+ = \sum_m C_{\mu m} a_m^+$

$$|e_{\mu}\rangle \rightarrow |e_m\rangle$$

$$\hat{A} = \sum_{\mu} A_{\mu\mu} a_{\mu}^+ a_{\mu}^- = \sum_{m,n} \overbrace{\left(\sum_{\mu} C_{\mu m} A_{\mu\mu} C_{\mu n}^* \right)}^{A_{mn}} a_m^+ a_n^-$$

$$\hat{A} = \sum_{m,n} A_{mn} a_m^+ a_n^-$$

$$\hat{A} | \dots n_{\mu} \dots n_{\nu} \dots \rangle = \sum_{\mu, \nu} A_{\mu\nu} a_{\mu}^+ a_{\nu}^- | \dots n_{\mu} \dots n_{\nu} \dots \rangle$$

$$= \sum_{\mu} \left(A_{\mu\mu} n_{\mu} | \dots n_{\mu} \dots \rangle + \sum_{\nu \neq \mu} A_{\mu\nu} \sqrt{n_{\nu}(n_{\nu}+1)} | \dots (n_{\mu}+1) \dots (n_{\nu}-1) \dots \rangle \right)$$

Two-Particle operators: (Coulomb interaction)

$$\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

$$\hat{B} = \sum_{i,j} B(\mathbf{q}_i, \mathbf{p}_i, \mathbf{q}_j, \mathbf{p}_j)$$

$$= \frac{1}{2} \sum_{\mathbf{k}\lambda, \mu\nu} \underline{B_{\mathbf{k}\lambda, \mu\nu}} a_{\mathbf{k}}^+ a_{\lambda}^+ \bar{a}_{\mu} \bar{a}_{\nu}$$

Normal ordering

$$B_{\mathbf{k}\lambda, \mu\nu} \stackrel{?}{=} \pm \langle e_{\mathbf{k}\lambda} | \hat{B} | e_{\mu\nu} \rangle$$

$$|e_{\mu\mu}\rangle = |\mu\rangle \otimes |\mu\rangle$$

$$|e_{\mu\nu}\rangle = \frac{|\mu\rangle \otimes |\nu\rangle \pm |\nu\rangle \otimes |\mu\rangle}{\sqrt{2}} \quad \mu \neq \nu$$

Proof:

$$\hat{B} |e_{\rho\sigma}\rangle = \hat{B} a_{\rho}^+ a_{\sigma}^+ |vac\rangle$$

$$= \frac{1}{2} \sum_{\mathbf{k}\lambda, \mu\nu} B_{\mathbf{k}\lambda, \mu\nu} a_{\mathbf{k}}^+ a_{\lambda}^+ \bar{a}_{\mu} \bar{a}_{\nu} a_{\rho}^+ a_{\sigma}^+ |vac\rangle$$

$$[\bar{a}_{\mu}, a_{\nu}^+]_{\pm} = \pm \delta_{\mu\nu} \rightarrow \bar{a}_{\mu} a_{\nu}^+ = a_{\nu}^+ \bar{a}_{\mu} \pm \delta_{\mu\nu} \mathbb{1}$$

$$\bar{a}_{\mu} \bar{a}_{\nu} a_{\rho}^+ a_{\sigma}^+ |vac\rangle = \bar{a}_{\mu} (a_{\rho}^+ \bar{a}_{\nu} \pm \delta_{\rho\nu} \mathbb{1}) a_{\sigma}^+ |vac\rangle$$

$$= \bar{a}_{\mu} a_{\rho}^+ \bar{a}_{\nu} a_{\sigma}^+ |vac\rangle + \delta_{\rho\nu} \bar{a}_{\mu} a_{\sigma}^+ |vac\rangle$$

$$= \pm \delta_{\mu\rho} \bar{a}_{\mu} a_{\rho}^+ |vac\rangle + \delta_{\rho\nu} \delta_{\mu\sigma} |vac\rangle$$

$$= (\delta_{\mu\sigma} \delta_{\nu\rho} \pm \delta_{\mu\rho} \delta_{\nu\sigma}) |vac\rangle$$

$$B |e_{\rho\sigma}\rangle = \frac{1}{2} \sum_{\kappa\lambda} \underbrace{(B_{\kappa\lambda,\sigma\rho} \pm B_{\kappa\lambda,\rho\sigma})}_{\pm 2 B_{\kappa\lambda,\rho\sigma}} a_{\kappa}^{\dagger} a_{\lambda}^{\dagger} |vac\rangle$$

$$= \pm \sum_{\kappa\lambda} B_{\kappa\lambda,\rho\sigma} a_{\kappa}^{\dagger} a_{\lambda}^{\dagger} |vac\rangle$$

$$B |e_{\rho\sigma}\rangle = \pm \sum_{\kappa\lambda} B_{\kappa\lambda,\rho\sigma} |e_{\kappa\lambda}\rangle$$

$$\langle e_{\kappa\lambda} | B |e_{\rho\sigma}\rangle = \pm B_{\kappa\lambda,\rho\sigma}$$

$$\hat{B} = \frac{1}{2} \sum_{\kappa\lambda\mu\nu} B_{\kappa\lambda,\mu\nu} a_{\kappa}^{\dagger} a_{\lambda}^{\dagger} \bar{a}_{\mu} \bar{a}_{\nu}$$

Quantum Fields:

$$\langle e_{\mu} | e_{\nu} \rangle = \int u_{\mu}^* u_{\nu} d^3\vec{r}$$

$$\underbrace{|e_{\mu}\rangle}_{\text{basis}} \rightarrow u_{\mu}(\vec{r}, t)$$

orthonormal: $\int u_{\mu}(\vec{r}, t)^* u_{\nu}(\vec{r}, t) d^3\vec{r} = \delta_{\mu\nu}$

Complete: $\sum_{\mu} u_{\mu}(\vec{r}', t)^* u(\vec{r}, t) = \delta(\vec{r} - \vec{r}')$

unitary transformation: $u_\mu(\vec{r}, t) = \sum_m C_{\mu m} u_m(\vec{r}, t)$

$$a_\mu^- = \sum_n C_{\mu n}^* a_n^-$$

$$\hat{\psi}(\vec{r}, t) := \sum_\mu u_\mu(\vec{r}, t) a_\mu^-$$

$$= \sum_{m, n} \left(\sum_\mu C_{\mu m} C_{\mu n}^* \right) u_m(\vec{r}, t) a_n^-$$

C : unitary $\rightarrow \delta_{mn}$

$$= \sum_m u_m(\vec{r}, t) a_m^- \rightarrow \hat{\psi} \text{ is invariant under unitary transformation}$$

$$\hat{\psi}(\vec{r}, t) = \sum_\mu u_\mu(\vec{r}, t) a_\mu^-, \quad \hat{\psi}^\dagger = \sum_\mu u_\mu^*(\vec{r}, t) a_\mu^+$$

linear PDE: if $u_\mu(\vec{r}, t)$ satisfy them $\rightarrow \hat{\psi}$'s also satisfy them

$$i\hbar \frac{\partial}{\partial t} u_\mu = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) u_\mu$$

$$i\hbar \frac{\partial}{\partial t} \hat{\psi} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \hat{\psi}$$

$$[\hat{\psi}(\vec{r},t), \hat{\psi}(\vec{r}',t)]_{\pm} = 0$$

$$[\psi(\vec{r},t), \psi^{\dagger}(\vec{r}',t)]_{\pm} = \delta(\vec{r}-\vec{r}')$$